Test Case #	Functional Requirement	Input Data/ Precondition	Expected Output/ Post Condition
1	The GUI only allows valid actions through visible controls/ buttons	The user attempts to interact with the interface.	The GUI restricts the user to available buttons, dropdown menus, and other controls, preventing invalid or mistyped actions
2	System shall allow the user to select a student type	The student types are provided in a dropdown menu, allowing the user to select from.	The selected student type will appear over the combobox.
3	Users must input a valid amount of credits completed (integer value > -1)	User can input any combination of valid data for creating the student alongside the credits completed and clicks the add student button Non-int: NN Negative: -1 Valid: 100	The system will reject non-integer or negative inputs for the credits completed and will display the appropriate error message. For the valid input, the user will have successfully added the user to the system (given the student has not been already added to the system)
4	Users must input a valid calendar date, meaning the date must exist, has already occurred, AND the student is older than 16.	Users can input any combination of data for creating a student. User enters DOB field and clicks add student button Non-existent date: 02/31/2000 Existing Date + <16: 02/28/2024 Valid Date + >16: 02/28/2003	The system will reject two first cases for adding a student, displaying the proper error message. As for the valid date, the user will successfully add the student to the system (given the student has not been already added to the system)

5	Users must input ALL of the required data to create a student (student type, first name, last name, dob, credits completed, and major).	All data must be valid entries (i.e. date, credits completed)	The system will alert the user that there are missing data tokens that need to be accounted for and it will prevent adding the student to the list.
6	User cannot add duplicate students (verified by first name, last name, and dob)	Student must already have been added with valid inputs and with the same identifying information (first name, last name, and dob) as the user's input (case-insensitive)	The system will prevent adding the duplicate student to the system and alert the user that the student already exists in the system.
7	The GUI shall provide state options when the student type is Tristate	The user selected Tristate from the Student Type dropdown	When Tristate is selected, NY and CT radio buttons become visible
8	User must select a state when student type is Tristate	The user selected Tristates as student type but doesn't select a state and adds student	The system will prevent adding the student to the list and will prompt the user to select a state.
9	User cannot select more than one state if the student is tristate	The user must select Tristate as the student type.	The system should remove the selection of one state if the other is selected.
10	State selection should reset when the student type changes	User must select tristate as the student type, a state, another student type, then revert back to tristate	The system should refresh the button group such that the state is not preselected as the user's last added tristate student.
11	The user can add a student	Given that the student does not already exist, the provided dob is a valid date, and the credits completed is a non-negative integer.	The system will add the student to the student list and update the user that the student has been created.
12	The system must properly load students.txt into the system after using the load	>> Load from file	The system will display all of the students from the .txt

	from file button		file being added to the list, as well as rejecting the students that already exist in the system as seen in a previous test case.
13	The system should notify the user if the student does not exist	The user enters identifying information for a student who is not in the list and clicks remove student	The system will alert the user if the student does not exist in the student list
14	The GUI should allow users to remove existing students	User enters identifying information of the student and clicks remove student	The student record is removed from the student list and the display updates accordingly.
15	The system prevents an invalid course number, time period, instructor, or classroom.	—	These values are chosen from combo boxes so invalid values cannot be entered.
16	A course cannot offer more than one section at the same time	A pre-existing section with the same course and time period as the user's input must already exist, irrespective of instructor and classroom.	The system will prevent a duplicate section from being offered, alerting the user that the course and period combination already exists in the schedule.
17	A course cannot be offered if there is a time conflict with the instructor	A pre-existing section with the instructor and time period the same as the user's input	The system will prevent the user from offering the section, alerting the user that there is a time conflict with the instructor.
18	A course cannot also cannot be offered if there is a time conflict with the room	A pre-existing section with the period and the classroom the same as the user's input	The system will prevent the user from offering the section, alerting the user that the room is not available during that time.

19	Users must provide ALL required information to offer a section (course, time, instructor, and classroom)	Partial information (For example: Course, Time, and Instructor)	The user will be alerted that more data is needed to create the section.
20	The user can create a section with a unique combination of course, period, instructor, and building	Given that there are no time conflicts for the instructor and building, and that section's period and course does not already exist	The schedule will update, adding the course alongside other offered course.
21	All enrollment fields must be filled before attempting enrollment.	The user leaves one ore more required enrollment fields blank and clicks the enroll button	The system will either alert the user that they are missing field(s)
22	When enrolling a student to a section, the target section must exist prior (i.e. already offered in schedule).	User enters a course and section that doesn't exist and enrolls	The system will prevent the user from enrolling the student to the section if it does not exist (i.e. was not offered prior) and an error message will appear.
23	Similarly, the target student must already exist in the student list when enrolling them to a section	The user attempts to enroll a student who is not in the student list	The system will prevent the user from enrolling the student to the section and alert the user that the student does not exist.
24	The user cannot enroll a student to a course twice	Both the student and section have to exist prior and the student must fulfill other aforementioned and following requirements	If the student is already enrolled in a section of that course, the system will prevent the user from enrolling the student to another section of that same course and alert the user.
25	The student of interest must fulfill the prerequirements of the course they are enrolling to.	Both the target student is registered and the section is offered.	If the student fulfills the requirements of the course they are enrolling to (major +

		The user inputs the identifying information of an existing student in the student list and the section's course and period in the schedule	standing), the system will **attempt** to enroll the student. Otherwise, the system will prevent enrolling the student to the section and alert the user what requirement the student is missing.
26	The user also cannot enroll a student to a section of a new course if there is a time conflict with the student (i.e. student is already enrolled in another section at the same time).	Both the target student is registered and the section is offered. The user inputs the identifying information of an existing student in the student list and the section's course and period in the schedule	If the student is already enrolled to another section during the period of the section of interest, the system will prevent enrolling the student. Otherwise, the system will **attempt** to enroll the student.
27	Students, when enrolling, will have their credits for the semester update while not exceeding their credit limits.	Both the target student is registered and the section is offered. The user inputs the identifying information of an existing student in the student list and the section's course and period in the schedule	If enrolling the student to the course will exceed their credit limit (based on their status), the system will prevent them from enrolling the student. Otherwise, the system will **attempt** to enroll the student.
28	A student cannot enroll in a section if the section is at max capacity.	The section has reached max capacity and the user attempts to enroll another student	An error message appears indicating that the section is full.
29	Students can enroll to a section	Given that the student fulfills the prereqs (major and standing), the section exists and has space in the roster, the credits for the semester will not exceed the	The system will update the roster to add that user's profile alongside other enrolled students.

		student’s credit limit, and there is no time conflict for the student	
30	When closing a section, the user’s section of interest (identified by course number and period) must be a valid, pre-existing section	—	If the section exists, the system will attempt to close the section Otherwise, the system will alert the user that the section does not exist.
31	The user cannot close a section if any student is enrolled to it	User inputs a valid course number and period of an existing section	If the roster of a section is not empty, the system will prevent closing the section and alert the user that at least one student is enrolled in the section.
32	Empty sections can be closed	The user selects an empty section and clicks the close section button	The system removed the section from the schedule
33	The system should prevent the user from dropping a non-existent student from a section	The user attempts to close a section that contains enrolled students	An error message appears indicating that the section cannot be closed.
34	The system should prevent the user from dropping non-enrolled students	The user attempts to drop a student who is not enrolled in the selected section	An error message appears indicating the student is not enrolled in the section.
35	Students can be dropped from sections	The user selects a student enrolled in a section and clicks the drop student button	The student is removed from the section roster.